

The Three-Rung Tower:

Mechanized Completeness for Intuitionistic, Gödel–Dummett, and Classical Higher-Order
Logic,
with Certified Evidence Readouts

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Abstract

We present a mechanized (Lean 4) development of completeness theorems for a single natural-deduction calculus for extensional Church-style higher-order logic at three strengths: the base calculus (intuitionistic), its extension by the prelinearity schema (Gödel–Dummett logic LC), and its extension by excluded middle (classical). All three completeness proofs share one Lindenbaum-style spine over one class of Heyting-valued substitutional models, and the containments between the three logics are themselves theorems, witnessed by small concrete countermodels. On top of the semantic core we build a certified *evidence readout* stack: proof-relevant derivation trees with provenance and cost semantics, monotone readouts from the derivability order into evidence carriers, and a separating-family theorem showing that evidence comparisons provably track derivability. A structural highlight is a *doctrine theorem*: the natural PLN evidence algebra is a complete Heyting algebra, but using it as the truth algebra of the logic forces the Gödel–Dummett axiom — so graded evidence enters intuitionistic reasoning as a certified readout, not as primitive truth values. Every theorem is stated with its live mechanization name and audited axiom footprint; the base soundness results and both canaries are choice-free.

1 Introduction

Reasoning systems that grade their conclusions — probabilistic logic networks, evidential and fuzzy reasoners, uncertain knowledge bases — need an answer to a foundational question: *what logic do the grades live over, and what guarantees that graded reports track the underlying entailment?* This paper answers the question for a mechanized higher-order logic by building the complete semantic ladder and then the certified bridges from proofs to grades.

The development has two halves.

The tower. One calculus — an intrinsically typed natural-deduction system for extensional Church-style higher-order logic, with no excluded middle built in — is proven sound and complete at three strengths with respect to one class of Heyting-valued models (Figure 1).

Read the tower bottom-up. On the bottom rung, truth means *verification*: a statement holds when it has been checked, and stays checked as knowledge grows — so $p \vee \neg p$ is not a law, because we may have verified neither disjunct (`em_not_derivable`, a two-world countermodel). On the middle rung, truth degrees behave like amounts of evidence: any two degrees are comparable, which is precisely the prelinearity axiom. On the top rung every statement receives a crisp verdict. Both containments are *strict as theorems*: a forked three-world frame refutes prelinearity intuitionistically

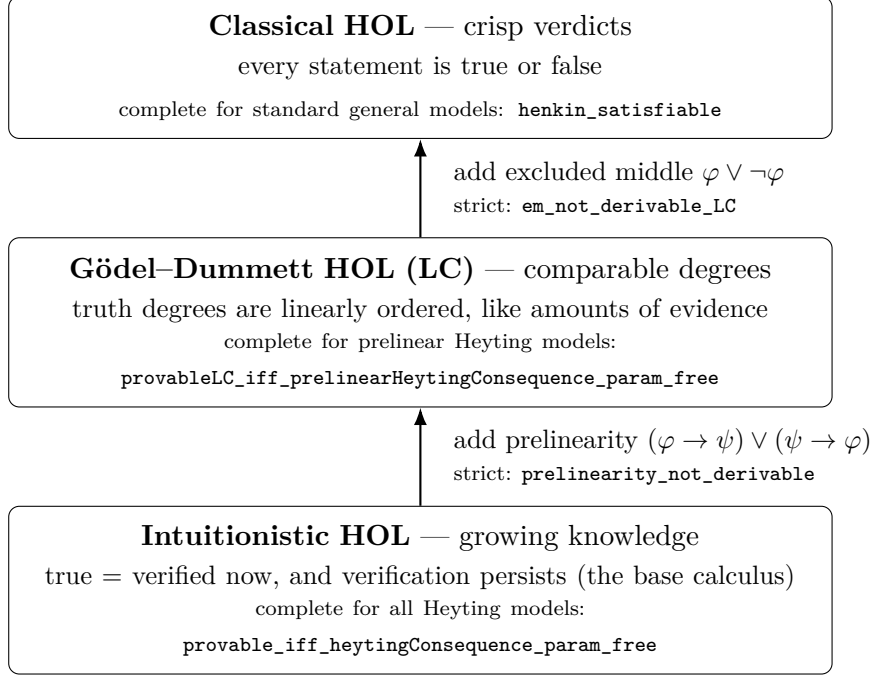


Figure 1: The three-rung tower: one calculus, one class of Heyting-valued models, three completeness theorems. Climbing a rung means adopting the axiom schema on the arrow; each climb is strict, witnessed by a concrete countermodel that lives on the lower rung and refutes the added schema.

(`prelinearity_not_derivable`), and a three-element Gödel chain refutes excluded middle inside LC (`em_not_derivable_LC`).

The readouts. The second half connects proofs to grades. Derivations are reified as proof-relevant trees with an axiom-free erasure to the propositional calculus; the same tree admits multiple semiring-style evaluations (a Boolean evaluation recovering derivability, source ledgers, proof-cost spectra, and evidence grades in the PLN carrier `BinaryEvidence`). Monotone readouts $\mu : \Omega \rightarrow Q$ from support orders into evidence carriers are axiomatized, and the completeness theorem itself is repackaged as a *separating family* of readouts: evidence comparisons across all theory-modelling evidence states jointly recover the derivability order (`heytingCrispSeparatingReadouts`). A doctrine theorem (Section 7) explains why the readout formulation is not merely convenient but forced.

Contributions.

1. Completeness for the EM-free extensional higher-order calculus with respect to Heyting-valued substitutional models, by a quotient-free Lindenbaum construction (Section 4).
2. Completeness for the Gödel–Dummett extension with respect to prelinear models, via a new schema-compatible fresh-parameter generalization obtained by abstracting whole derivations (Section 5).
3. Strictness of the tower, by three small concrete countermodels built with a reusable logical-relations recipe (Section 6).

4. The doctrine theorem: the PLN evidence algebra `BinaryEvidence` is a frame whose Heyting structure validates prelinearity, so evidence-as-truth-values is Gödel–Dummett logic, not intuitionistic logic (Section 7).
5. A certified readout stack from proof-relevant derivations to provenance, cost, and evidence grades, with a separating-family completeness repackaging and first exactness certificates for inference rules (Section 8).

All results are mechanized in Lean 4 inside the MeTTapedia development. We write theorem names in typewriter font; Table 1 lists axiom footprints. Soundness of the base calculus and both negative canaries use only `propext` and `Quot.sound` (no choice); the completeness directions additionally use `Classical.choice`, as expected of Henkin-style constructions.

2 The calculus

Terms are intrinsically typed: types are generated from a set of base types and a distinguished proposition type by the arrow, and a single inductive family of terms covers the λ -calculus fragment (variables, constants, application, abstraction) and the logical constructors ($\top, \perp, \wedge, \vee, \rightarrow, \neg, =, \forall, \exists$), formulas being terms of proposition type. The signature is extended by a countable supply of *parameters* at every type; parameters act as fresh constants for witnessing and generalization.

The proof system `ExtDerivation` is natural deduction over lists of hypotheses with the standard introduction/elimination rules, plus an extensional equality package: reflexivity, symmetry, transitivity, application congruence in both positions, ξ (equality under λ), functional extensionality, β , η , and propositional extensionality (mutually implying propositions are equal). Excluded middle is *not* a rule; classical strength enters only as an axiom set. Derivability from a theory T (a set of closed formulas), written $\text{Prov}_T(\theta)$, means derivability from a finite list of members of T .

Two structural facts drive everything later: the deduction theorem (an equivalence between $\text{Prov}_{T \cup \{\varphi\}}(\psi)$ and $\text{Prov}_T(\varphi \rightarrow \psi)$), and fresh-parameter generalization: if $\varphi[c]$ is derivable and the parameter c occurs in neither the hypotheses nor φ , then $\forall x.\varphi$ is derivable (`provable_all_intro_fresh`).

3 Heyting-valued substitutional models

Definition 3.1 (`HeytingGeneralModel`). A model consists of a preordered value carrier Ω with distinguished $\top, \perp$, binary $\sqcap, \sqcup$, and a Heyting implication $\Rightarrow$ satisfying the residuation laws and finite distributivity, together with a valuation val on closed formulas such that: the propositional connectives are interpreted by the corresponding operations (as equations); negation is sandwiched between $\text{val}(\varphi) \Rightarrow \perp$ from both sides; the quantifiers satisfy *substitutional bounds and exactness* — $\text{val}(\forall x.\varphi)$ is the greatest lower bound of the values of all closed-term instances, and dually for $\exists$; and the equality rules hold as inequalities ($\top \leq \text{val}(t = t)$, transport along application, ξ , functional and propositional extensionality, β , η).

The class is deliberately minimal: no completeness of the lattice is assumed, only the specific formula-indexed bounds the quantifier rules require. Complete Heyting algebras, frames of upsets of a preorder, and the Lindenbaum structure below are all instances.

Theorem 3.2 (Soundness; `HeytingSem.sound`). *Every derivation is valid in every model: if $\Delta \vdash \varphi$ then for every model, every closing substitution ρ , the meet of the hypothesis values under ρ lies below $\text{val}(\varphi\rho)$. Axioms: `propext`, `Quot.sound`.*

```

-- Mettapedia/Logic/HOL/Semantics/HeytingGeneral.lean (namespace HeytingSem)
theorem sound {Γ : Ctx Base} {Δ : List (Formula Const Γ)} {φ : Formula Const Γ}
  (d : ExtDerivation Const Δ φ) :
  ∀ (M : HeytingGeneralModel.{u, v, w} Base Const) (ρ : Subst Const Γ []),
  M.le (M.hypVal (Δ.map (subst ρ))) (M.val (subst ρ φ))

```

The proof is a single induction over all twenty-eight rules; the quantifier cases ride one substitution-valuation commutation lemma, and the extensionality cases use only the inequational equality package.

Kripke models embed: every substitutional Kripke-style model induces a Heyting-valued model on the upsets of its frame (ofKripkeHenkin; axioms `propext`, `Quot.sound`), so the familiar possible-worlds reading is a special case of the algebraic one.

4 Intuitionistic completeness

The canonical model is Lindenbaum-style but *quotient-free*: since the class only demands a preorder, the value carrier is the set of closed formulas itself, the valuation is the identity, and

$$\varphi \leq \psi \quad :\Longleftrightarrow \quad \text{Prov}_{T \cup \{\varphi\}}(\psi).$$

All valuation equations hold by reflexivity; the lattice and residuation laws are the deduction theorem and short natural-deduction compositions; and quantifier *exactness* — any lower bound of all instances lies below the universal — is exactly fresh-parameter generalization: if $\text{Prov}_{T \cup \{\omega\}}(\varphi[t])$ for every closed t , instantiate at a parameter c fresh for ω and φ (possible since T is parameter-free and the relevant formulas mention finitely many parameters) and generalize.

Theorem 4.1 (Intuitionistic completeness; `provable_iff_heytingConsequence_param_free`).
For parameter-free theories T over the parameter-extended signature and closed θ :

$$\text{Prov}_T(\theta) \Longleftrightarrow \text{every Heyting-valued model that top-values } T \text{ top-values } \theta.$$

Axioms: `propext`, `Classical.choice`, `Quot.sound`.

```

-- Mettapedia/Logic/HOL/Semantics/HeytingGeneral.lean
def HeytingConsequence (T : ClosedTheorySet Const) (θ : ClosedFormula Const) : Prop :=
  ∀ M : HeytingGeneralModel.{u, v, max u v} Base Const,
  (∀ ψ ∈ T, M.le M.top (M.val ψ)) → M.le M.top (M.val θ)

theorem provable_iff_heytingConsequence_param_free
  {T : ClosedTheorySet (WithParams Const)}
  {θ : ClosedFormula (WithParams Const)}
  (hT0 : ∀ ψ ∈ T, ∀ (σ : Ty Base) (k : Nat),
    NoConstOccurrence (param σ k : WithParams Const σ) ψ) :
  ClosedTheorySet.Provable (Const := WithParams Const) T θ ↔
  HeytingConsequence (Base := Base) T θ

```

Remark 4.2 (Why not a truth lemma). A more traditional route — a canonical Kripke model with a structural truth lemma — runs into the impredicativity obstruction familiar from the Takeuti problem: in higher-order logic, instantiating a propositional variable can produce a formula more complex than the quantified formula, so valuations do not extend by induction on formulas, and the natural canonical-instance requirements become provably unsatisfiable for compositional models. This obstruction is faithfully recorded in the mechanization, and following DeMarco and Lipton’s

analysis [3], the Lindenbaum term model (available because the calculus proves its own cut) is the sanctioned bypass. The canonical Kripke machinery survives as a separate presentation-layer bridge, conditional on an enumeration package exactly analogous to the enumeration hypotheses of the classical Henkin proof.

Theorem 4.3 (The canary; `em_not_derivable`). $\not\vdash p \vee \neg p$ for a propositional constant p . Axioms: *propext*, *Quot.sound* — no choice.

The countermodel is the two-world growing-knowledge frame ($\circ \rightarrow \bullet$, p true only at the later world), built as a Heyting-valued model over full function spaces with a hereditarily extensional logical relation as the admissibility predicate; the fundamental lemma of that relation discharges the model obligations without any normalization theory. The same recipe (frame + logical relation + fundamental lemma) is reused for both strictness canaries below.

5 The Gödel–Dummett rung

Gödel–Dummett logic LC extends intuitionistic logic by *prelinearity*: $(\varphi \rightarrow \psi) \vee (\psi \rightarrow \varphi)$. We do not change the calculus; we extend theories:

$$\text{Prov}_T^{\text{LC}}(\theta) := \text{Prov}_{T \cup \text{lcSchema}}(\theta),$$

where the schema contains all closed prelinearity instances *and all their universal closures of every depth* (an inductive shape `PrelinShape`: prelinearity instances closed under $\forall$). The closures are not decoration; they are what makes the following crux go through.

Lemma 5.1 (Schema-compatible generalization; `lc_all_intro_of_instances`). *Over $T \cup \text{lcSchema}$ with T parameter-free, if every closed instance of φ is derivable (uniformly in a side formula ω), then $\forall x.\varphi$ is derivable.*

The difficulty is that schema members mention every parameter, so no parameter is fresh for the whole theory and the classical generalization lemma does not apply. The mechanized solution abstracts the *whole witnessing derivation* over the chosen parameter c (`abstractConstAt_deriv`): parameter-free hypotheses abstract to their weakenings; schema hypotheses abstract to *open* prelinearity shapes, which are recovered inside the extended context from their universal closures — schema members again — by instantiating at the newly bound variable; a generic hypothesis-substitution combinator (`ExtDerivation.cut_hyps`) reassembles the derivation, and ordinary $\forall$ -introduction concludes. No renaming machinery and no proof-theoretic detour is needed.

With the crux in place, the Lindenbaum construction is reused verbatim: the model constructor is parameterized by the generalization capability (`lindenbaumModelOfAllIntro`), the intuitionistic model instantiates it with the classical lemma, the LC model with the crux; prelinearity of the LC model is schema membership.

Theorem 5.2 (LC completeness). *For parameter-free T : $\text{Prov}_T^{\text{LC}}(\theta)$ iff θ is top-valued in every pre-linear Heyting-valued model that top-values T . Axioms: *propext*, *Classical.choice*, *Quot.sound*.*

```
-- Mettapedia/Logic/HOL/Semantics/GoedelDummett.lean
inductive PrelinShape : {Γ : Ctx Base} → Formula (WithParams Const) Γ → Prop
| base {Γ : Ctx Base} (A B : Formula (WithParams Const) Γ) :
  PrelinShape (.or (.imp A B) (.imp B A))
| all {Γ : Ctx Base} {σ : Ty Base} {body : Formula (WithParams Const) (σ :: Γ)} :
  PrelinShape body → PrelinShape (.all body)
```

```

def lcSchema (Const : Ty Base → Type v) : ClosedTheorySet (WithParams Const) :=
  { χ | PrelinShape (Base := Base) χ }

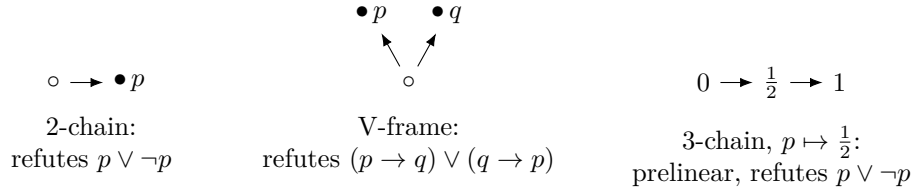
def ProvableLC (T : ClosedTheorySet (WithParams Const))
  (θ : ClosedFormula (WithParams Const)) : Prop :=
  ClosedTheorySet.Provable (Const := WithParams Const)
  (T ∪ lcSchema (Base := Base) Const) θ

def PrelinearHeytingConsequence (T : ClosedTheorySet (WithParams Const))
  (θ : ClosedFormula (WithParams Const)) : Prop :=
  ∀ M : HeytingGeneralModel.{u, v, max u v} Base (WithParams Const),
  (∀ a b : M.Ω, M.le M.top (M.sup (M.himp a b) (M.himp b a))) →
  (∀ ψ ∈ T, M.le M.top (M.val ψ)) → M.le M.top (M.val θ)

theorem provableLC_iff_prelinearHeytingConsequence_param_free
  {T : ClosedTheorySet (WithParams Const)}
  {θ : ClosedFormula (WithParams Const)}
  (hT0 : ∀ ψ ∈ T, ∀ (σ : Ty Base) (k : Nat),
    NoConstOccurrence (param σ k : WithParams Const σ) ψ) :
  ProvableLC (Base := Base) T θ ↔
  PrelinearHeytingConsequence (Base := Base) T θ

```

6 Strictness



Theorem 6.1 (Strict containments).

- *intuitionistic_strictly_below_lc*: prelinearity is not derivable in the EM-free calculus (*prelinearity_not_derivable*, the V-frame countermodel), while it is LC-derivable as a schema member (*provableLC_prelinearity*; axioms *propext* only).
- *lc_strictly_below_classical*: excluded middle is not LC-derivable — the three-element Gödel chain is prelinear and refutes it at the middle value (*em_not_derivable_LC*) — while it is classically derivable.

```

-- Mettapedia/Logic/HOL/Semantics/KripkeHenkinCountermodel.lean
theorem em_not_derivable :
  ¬ ClosedTheorySet.Provable (Const := EMCanaryConst)
  (∅ : ClosedTheorySet EMCanaryConst) emCanaryExcludedMiddle

-- Mettapedia/Logic/HOL/Semantics/ForkedFrameCountermodel.lean
theorem prelinearity_not_derivable :
  ¬ ClosedTheorySet.Provable (Const := ForkedFrameConst)
  (∅ : ClosedTheorySet ForkedFrameConst) forkedPrelinearity

-- Mettapedia/Logic/HOL/Semantics/GoedelDummettCountermodel.lean
theorem intuitionistic_strictly_below_lc :

```

```

(¬ ClosedTheorySet.Provable (Const := ForkedFrameConst)
  (∅ : ClosedTheorySet ForkedFrameConst) forkedPrelinearity) ∧
ProvableLC (Base := ForkedFrameBase) (Const := ForkedFrameConst)
  (∅ : ClosedTheorySet (WithParams ForkedFrameConst)) forkedPrelinearityLC

theorem lc_strictly_below_classical :
  (¬ ProvableLC (Base := EMCanaryBase) (Const := EMCanaryConst)
    (∅ : ClosedTheorySet (WithParams EMCanaryConst)) emCanaryExcludedMiddleLC) ∧
  ClosedTheorySet.Provable (Const := WithParams EMCanaryConst)
    (EMSchema (Base := EMCanaryBase) EMCanaryConst) emCanaryExcludedMiddleLC

```

Together with the classical Henkin completeness previously mechanized in the same development (`henkin_satisfiable`), the tower is complete: three completeness theorems, two strict containments, one calculus, one model class.

7 The doctrine theorem: evidence is a readout

The PLN evidence carrier `BinaryEvidence` consists of pairs $(n^+, n^-) \in [0, \infty]^2$ of positive and negative evidence. Under the coordinatewise order it is a *frame* — a complete Heyting algebra with a Gödel-style implication computed componentwise — so evidence-valued models of the calculus genuinely exist. But each coordinate is a chain, and products of chains are prelinear:

Theorem 7.1 (Evidence-as-truth forces prelinearity). *$(a \Rightarrow b) \sqcup (b \Rightarrow a) = \top$ holds in `BinaryEvidence` (`binaryEvidence_goedelDummett`), while excluded middle fails at the balanced value $(1, 1)$ (`binaryEvidence_not_boolean`). Consequently every `BinaryEvidence`-valued model of the calculus — equivalently, every pushforward of a model along a structure-preserving map into `BinaryEvidence`, `HeytingGeneralModel.pushforward` — validates the LC schema (`pushforward_binaryEvidence_validates_lc`).*

```

-- Mettapedia/PLN/Bridges/HOL/EvidenceValuedModels.lean
theorem binaryEvidence_goedelDummett (a b : BinaryEvidence) :
  (a ⇒ b) ∪ (b ⇒ a) = ⊤

theorem binaryEvidence_not_boolean :
  ∃ a : BinaryEvidence, a ∪ (a ⇒ ⊥) ≠ ⊤

```

Since prelinearity is *not* intuitionistically derivable (Section 6), taking evidence values as primitive truth values silently strengthens the logic to Gödel–Dummett. This is the precise algebraic form of the support/evidence separation: intuitionistic truth lives in Heyting support; graded evidence enters as a monotone *readout* of it; and if a graded, prelinear semantics is wanted on purpose, the LC rung of the tower is its certified home. The doctrine is thus not a design taste but a theorem, and the tower supplies certified semantics on both sides of it.

8 Certified evidence readouts

Readout interfaces. A readout is a monotone map $\mu : \Omega \rightarrow Q$ from a support order into an evidence carrier (`WMReadout`); an order-reflecting readout recovers support from evidence; a *separating family* $\{\mu_i\}$ jointly recovers it: $(\forall i. \mu_i a \leq \mu_i b) \Rightarrow a \leq b$ (`SeparatingWMReadouts`).

Theorem 8.1 (Completeness as separation). *The pointed evidence states (M, ω) — a model with a distinguished value, satisfaction being $\omega \leq \text{val}(\varphi)$ — that model a parameter-free theory T , reading*

formulas to crisp singleton query strengths, form a separating family for the derivability order $\varphi \leq \psi : \Leftrightarrow \text{Prov}_T(\varphi \rightarrow \psi)$. In particular $T \vdash \varphi \rightarrow \psi$ iff the strength of φ is at most that of ψ at every such state.

```
-- Mettapedia/PLN/Bridges/HOL/PLNHeytingHOLWorldModelBridge.lean
theorem provable_imp_iff_singletonStrengthLEOnTheory
  {T : ClosedTheorySet (WithParams Const)}
  (hT0 : ∀ ψ ∈ T, ∀ (σ : Ty Base) (k : Nat),
    NoConstOccurrence (param σ k : WithParams Const σ) ψ)
  (φ ψ : ClosedFormula (WithParams Const)) :
  ClosedTheorySet.Provable (Const := WithParams Const) T (.imp φ ψ) ↔
    singletonStrengthLEOnTheory (Base := Base) T φ ψ

noncomputable def heytingCrispSeparatingReadouts
  (T : ClosedTheorySet (WithParams Const))
  (hT0 : ∀ ψ ∈ T, ∀ (σ : Ty Base) (k : Nat),
    NoConstOccurrence (param σ k : WithParams Const σ) ψ) :
  Mettapedia.PLN.WorldModel.SeparatingWMReadouts
  {S : PointedHeytingHOL Base (WithParams Const) // stateModelsTheory T S}
  (ClosedFormula (WithParams Const)) ℝ≥0∞
  (fun φ ψ => ClosedTheorySet.Provable (Const := WithParams Const) T (.imp φ ψ))
  (· ≤ ·)
-- (instance fields mu / monotone / separating: proofs in source)
```

The completeness direction is carried by the Lindenbaum model *pointed at the antecedent itself* — the canonical model’s states are formulas, so the separating family is definable without any further construction. This is the faithfulness guarantee for graded reporting: evidence comparisons provably track derivability and cannot silently diverge from the logic.

Proof-relevant derivations and their evaluations. A `DerivationTree` mirror of the calculus (with an axiom-free erasure and `nonempty_iff_extDerivation`) supports semiring-style evaluation: the Boolean evaluation recovers derivability (`boolTreeProvable_iff_provable`); source ledgers (`sourceIdeal`) record which hypotheses a proof used; cost spectra (`costSpectrum`, `costUpperBounds`) grade proof effort, with monotone transport along derivable implication up to an explicit additive slack; and evidence grades (`evGrade`) land in `BinaryEvidence`, aggregating over finite bags of source-disjoint derivations by evidence addition, so “more independent derivations, more evidence” is a theorem. The same derivation object thus admits a trust-audit reading and a graded reading — the provenance-semiring perspective [8] applied to proof theory.

Exactness certificates. For inference rules, the readout stack supports *exactness* statements: the evidence transform effected by a rule application either equals a closed-form combination of the premise grades or is honestly bounded by one, with exhibited lossy cases. The first such certificate, for modus ponens, is an equality (`evGrade_impE_exact`); the programme of regrounding the PLN rule families (deduction, revision, induction/abduction) as exactness or bound certificates against this semantics is under way, with revision already half-captured by the source-disjoint aggregation law.

```
-- Mettapedia/PLN/Bridges/HOL/BinaryEvidenceReadout.lean
theorem evGrade_impE_exact
  (dImp : DerivationTree Const Δ (.imp φ ψ))
  (dφ : DerivationTree Const Δ φ) :
  evGrade (DerivationTree.impE dImp dφ) =
    ruleRevisionCandidate (evGrade dImp) (evGrade dφ)
```

9 Axiom audit

Theorem	Axioms
HeytingSem.sound	propext, Quot.sound
em_not_derivable	propext, Quot.sound
ofKripkeHenkin	propext, Quot.sound
provableLC_prelinearity	propext
provable_iff_heytingConsequence_param_free	propext, Classical.choice, Quot.sound
provableLC_iff_prelinearHeytingConsequence_param_free	propext, Classical.choice, Quot.sound
binaryEvidence_goedelDummett	propext, Classical.choice, Quot.sound
provable_imp_iff_singletonStrengthLEOnTheory	propext, Classical.choice, Quot.sound

Table 1: Axiom footprints (`#print axioms`). Soundness and the negative canaries are choice-free; completeness directions use classical choice, as expected of Henkin-style constructions.

10 Related work

Henkin’s completeness for the theory of types [1] and Andrews’s general-models tradition [2] supply the classical rung and the general-model discipline used throughout. DeMarco and Lipton [3] and Hermant and Lipton [4] develop completeness and cut-elimination for intuitionistic type theory and identify the impredicative obstruction to truth-lemma-style arguments that our mechanization reproduces and routes around; our Lindenbaum shortcut corresponds to their with-cut term-model observation. Gödel’s finite-valued systems [5] and Dummett’s axiomatization of LC [6] ground the middle rung; the algebraic completeness tradition for such logics goes back to Rasiowa and Sikorski [7], whose enough-constants quantifier treatment is exactly the fresh-parameter exactness used here. Provenance semirings are due to Green, Karvounarakis, and Tannen [8]. Mechanized completeness proofs for first-order and propositional intuitionistic systems (e.g. [9, 10]) informed the enumeration-hypothesis discipline of the canonical bridge; to our knowledge the mechanized Gödel–Dummett completeness at higher order, the strictness tower, and the doctrine theorem are new.

11 Conclusion and outlook

One calculus, one model class, three certified strengths, strict containments, and a graded readout stack whose faithfulness to derivability is itself a theorem. Open continuations, each scoped in the development: completeness of the LC rung with respect to **BinaryEvidence**-valued models specifically (a quantifier-aware subdirect representation into chain products); an enumeration-package instance unconditionalizing the canonical Kripke bridge; the systematic regrounding of graded inference-rule families as exactness certificates; and the agent-facing side — using the certified readouts as an introspection layer in which a reasoning system’s confidence reports are recomputed from proof certificates rather than stored as unaccountable numbers, with retraction handled provenance-exactly. The tower makes those reports trustworthy by construction: whatever an agent says about the strength of its beliefs, the mathematics underneath says it about derivability too.

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